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Improved Event-Triggered Adaptive Neural Network Control for Multi-agent Systems Under Denial-of-Service Attacks

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ABSTRACT

This paper addresses the consensus problem of nonlinear multi-agent systems subject to external disturbances and uncertainties under denial-of-service (DoS) attacks. Firstly, an observer-based state feedback control method is employed to achieve secure control by estimating the system's state in real time. Secondly, by combining a memory-based adaptive event-triggered mechanism with neural networks, the paper aims to approximate the nonlinear terms in the networked system and efficiently conserve system resources. Finally, based on a two-degree-of-freedom model of a vehicle affected by crosswinds, this paper constructs a multi-unmanned ground vehicle (Multi-UGV) system to validate the effectiveness of the proposed method. Simulation results show that the proposed control strategy can effectively handle external disturbances such as crosswinds in practical applications, ensuring the stability and reliable operation of the Multi-UGV system.

Multi-agent systems are confronted with significant challenges in complex and open environments[1], including limited environmental perception capabilities[2], insufficient autonomy delays in communication[3], and vulnerability to cyber attacks[4]. These issues have garnered increasing attention from scholars, who are actively exploring solutions through various approaches such as security control[5], formation control[6], and consensus control[7], among others. Among the above, security control stands out as a critical and highly prioritized field, especially in applications such as unmanned workshops, unmanned vehicles[8], unmanned ships[9], unmanned power grids, robots and other fields[10].

To address the challenges posed by cyber attacks on control systems, X. CAI *et al.* have designed an enhanced data compression mechanism to effectively mitigate the impact of denial-of-service (DoS) attacks on nonlinear networked control systems[11]. In a related effort, L. W. AN *et al.* have proposed a compensation control method that utilizes estimated bounds of attack uncertainty rather than relying on exact values[12]. Furthermore, scholars have developed a resilient controller based on attack estimation, eliminating the need for control reconstruc-

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tion[13]. Z. H. CHENG *et al.* have investigated the consensus problem of event-triggered control for linear multi-agent systems under periodic DoS attacks, proposing sufficient conditions based on linear matrix inequalities (LMIs) to ensure exponential consensus of the system[14]. These studies have introduced new methods and strategies for fault-tolerant control under cyber attacks, aiming to address the challenges in cooperative problems. However, the DoS attacks in Ref. [14] have been modeled as periodic events, whereas in real-world scenarios, such attacks typically occur at random intervals. Therefore, to address this limitation, this paper proposes an observer-based state feedback control approach for achieving secure control of the system under randomly initiated DoS attacks.

For multi-agent nonlinear systems, studies such as Ref. [15-16] have used the Lipschitz continuity condition to handle the nonlinear terms in multi-agent systems and have employed Lyapunov functions to achieve consensus. For nonlinear control problems in mechanical systems, Ref. [17] has adopted small-angle approximation methods to linearize the system, making it easier for the controller to achieve stability and performance objectives. Furthermore, C. C. HUA *et al.* have proposed an adaptive fuzzy control method to address the random disconnection nonlinear issue in multi-agent systems, offering a robust solution to this complex issue[18]. In addition, H. LI *et al.* have innovatively proposed a control method based on intermittent output feedback, effectively addressing the nonlinear issues of the system[19]. L. Y. SHI *et al.* have presented a modeling approach based on the semi-Markov process to solve the bipartite synchronization problem for a class of continuous-time nonlinear cooperation-competition networks[20]. Notably, all of the aforementioned studies rely on certain assumptions regarding the nonlinear terms in multi-agent systems. In this study, we employ neural network (NN) technology to estimate the nonlinear terms, without relying on any assumptions, thereby providing a more general and practical solution for real-world applications.

Since both the information transmission tasks between the modules of the networked system and the computational tasks of each module rely on the system's network channels, the modules compete with each other for access to the limited network bandwidth resources[21-22]. Therefore, reducing resource waste has also been a key issue addressed in the present work. Analysis of the existing literature has revealed that Ref. [23] has addressed the issue of communication resource conservation in robotic manipulator systems through a dynamic event-triggered mechanism. In Ref. [24], the scholars have introduced dynamic variables to achieve fault-tolerant synchronization in the system and have significantly reduced the transmission frequency of state information in network communication. S. YUAN *et al.* have proposed an adaptive event-triggered control strategy, where the threshold parameters have been adjusted adaptively based on system performance requirements, effectively addressing the security consensus issue in delay-prone multi-agent systems under non-periodic DoS attacks in Ref. [25]. Furthermore, the memory event-triggered mechanism has been developed to assign varying weights to historical data, achieving greater system resource savings compared to the methods proposed in Ref. [26]. By utilizing a memory-based event-triggered mechanism combined with the Lipschitz inequality, the security consensus problem for nonlinear multi-agent systems under periodic denial-of-service attacks has been addressed in Ref. [27]. Inspired by the above literature, the main contributions of this paper are as follows:

1) This paper employs an observer-based attack-resilient security control approach to address the DoS attacks that frequently occur in real-world applications;

2) A method combining memory-based NNs and adaptive event-triggered mechanisms is used to approximate the nonlinear terms of the system, which are then incorporated into the controller design. This approach aims to enhance system stability and improve resource efficiency.

The remaining 4 sections of this paper are organized as follows: Section 1 provides a detailed explanation of the fundamental theoretical knowledge and the establishment of the system model, clarifying the research object. Section 2 presents the main stability theorems and the designed controllers. Section 3 verifies the effectiveness of the proposed method through simulations and experiments. Finally, Section 4 summarizes the paper, highlighting its limitations and suggesting directions for future research.

Notations To facilitate the subsequent analysis and derivation, the following symbols used in this paper are defined: $\boldsymbol{R}^p$ represents all $p\times 1$ real matrices. $\boldsymbol{I}_N$ and $\boldsymbol{1}_N$ represent the N-dimensional identity matrix and the N-dimensional vector with all elements equal to 1, respectively. The notations $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ are the maximum and minimum eigenvalues of a matrix, respectively. He($\boldsymbol{A}$) denotes a shorthand of $\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}}$. And each agent is regarded as a node, and the undirected graph $\mathcal{G}=\{\mathcal{V},\mathcal{E},\mathcal{A}\}$ is used to represent the topology, where the notation $\mathcal{V}=\{1,2,\cdots,m\}$ denotes the set of nodes, $\mathcal{E}\subseteq\{\mathcal{V}\times\mathcal{V}\}$ represents the set of edges, and the $\mathcal{A}=[a_{ij}]_{m\times m}$ represents the communication weights between agents, with $a_{ii}=0$. If $(j,i)\in\mathcal{E}$, then $a_{ij}=1$; otherwise, $a_{ij}=0$. Furthermore, the notation $\mathcal{N}_i=\{j\in\mathcal{V}\mid(j,i)\in\mathcal{E},j\neq i\}$ denotes the neighbor set of agent i. And $\mathcal{L}=[l_{ij}]_{m\times m}$ is the Laplacian matrix, satisfying $l_{ii}=\sum_{j\in\mathcal{N}_i}a_{ij}$ (while $i=j$) and $l_{ij}=-a_{ij}$ (while $i\neq j$). * denotes the corresponding symmetric part in the symmetric matrix.

1 System Formulation and Preliminaries

The detailed information on the considered nonlinear multi-agent systems, basic assumptions and lemmas, event-triggered schemes, and the adaptive NN-based control design problem will be stated in this section.

1.1 System Description

Consider the following class of Multiple Input Multiple Output (MIMO) high-order nonlinear multi-agent systems:

$$\begin{cases} \dot{\boldsymbol{x}}_i(t) = (\boldsymbol{A}+\Delta\boldsymbol{A}(t))\boldsymbol{x}_i(t)+\boldsymbol{B}\bar{\boldsymbol{u}}_i(t)+ \\ \qquad \boldsymbol{D}\boldsymbol{d}_i(t)+\boldsymbol{F}\boldsymbol{g}_i(\boldsymbol{x}_i(t)) \\ \boldsymbol{y}_i(t) = \boldsymbol{C}\boldsymbol{x}_i(t) \qquad\qquad i = 1,\cdots,N \end{cases} \tag{1}$$

where $\boldsymbol{x}_i(t) \in \boldsymbol{R}^p$ and $\bar{\boldsymbol{u}}_i(t) \in \boldsymbol{R}^q$ are the state vector and control input of the ith subsystem, respectively; $\boldsymbol{d}_i(\boldsymbol{t})$ represents the external disturbance generated by the exogenous system with the following parameter matrix, and it is bounded, *i.e.*, $\|\boldsymbol{d}_i(t)\| < d_r$, where d_r is a given known positive constant representing the maximum value of the disturbance $\boldsymbol{d}_i(t)$; $\boldsymbol{g}_i(\boldsymbol{x}_i(t))$ is a nonlinear function of the state variable $\boldsymbol{x}_i(t)$ with an arbitrary form; $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, $\boldsymbol{D}$ and $\boldsymbol{F}$ are the constant state matrices; the matrix $\Delta\boldsymbol{A}(t)$ is a time-varying uncertainty.

The following basic assumptions, definitions and lemma are used to present the main results.

Assumption 1 The system matrices $(\boldsymbol{A},\boldsymbol{B})$ are stabilizable, and the system matrices $(\boldsymbol{A},\boldsymbol{C})$ are detectable in Eq. (1).

Assumption 2[27] The uncertainty $\Delta\boldsymbol{A}(t)$ in system (1) is assumed to satisfy:

$$\Delta\boldsymbol{A}(t) = \tau\boldsymbol{A}\boldsymbol{E}(t)$$

where τ is an uncertain parameter and $\boldsymbol{E}(t)$ represents an unknown time-varying matrix, satisfying $\boldsymbol{E}^{\mathrm{T}}(t)\boldsymbol{E}(t) \leqslant \boldsymbol{I}_N$.

Definition 1 The considered multi-agent system (1) with a communication topology represented by an undirected graph can reach consensus in the absence of a leader, if the following equation holds:

$$\lim_{t\to\infty}\left\|\boldsymbol{x}_i(t) - \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{x}_i(t)\right\| = 0$$

Definition 2[28] For given positive constants c and χ, the initial state satisfying:

$$\|x_{t_0}\| < \vartheta < c$$

for any initial perturbation ϑ, then the system is said to be Uniformly Ultimately Bounded (UUB), if there exists a time $T(\vartheta,\chi)$ such that the state of the system remains within:

$$\|x_t\| < \chi$$

for all $t \geqslant t_0 + T$.

Lemma 1[29] Let $\psi(t) \in L_\infty$ be a real-valued function. If there exists a positive definite function $V(t)$ with $V(0)$ bounded and satisfying the following inequality:

$$\dot{V}(t) \leqslant -l_1 V(t) + l_2\psi(t) \quad \forall t \in \mathbb{R}^+$$

where $l_1 > 0$ and l_2 are constants, then the function $V(t)$ is bounded.

1.2 Adaptive Neural Network Function Approximation

Different from Ref. [27], the smooth nonlinear function $g(\cdot)$ is approximated by using the NN technique mentioned in Ref. [30]. In this paper, the nonlinear term $g_i(\boldsymbol{x}_i(t))$ in the considered system (1) is approximated by using a Radial Basis Function Neural Network (RBFNN), given as follows:

$$\boldsymbol{g}_i(\boldsymbol{x}_i(\boldsymbol{t})) = \boldsymbol{\Theta}_i^{\mathrm{T}}\boldsymbol{h}_i(\boldsymbol{x}_i) + \varepsilon_i \quad \forall \boldsymbol{x}_i \in \Omega \tag{2}$$

where $\boldsymbol{\Theta}_i(t) \in \boldsymbol{R}^p$ represents the weight matrix of the ideal NN, which satisfies $\|\boldsymbol{\Theta}_i\| \leqslant \Theta_G$; $\boldsymbol{h}_i(\boldsymbol{x}_i) = [h_{i1}\ h_{i2}\ \cdots\ h_{in}]^{\mathrm{T}}$ represents the output of the Gaussian function, which is bounded and satisfies $h_{il}(x_i) = \mathrm{e}^{\frac{-\|x-\iota_l\|^2}{b_l^2}}$ with $l = 1,\cdots,n$ and $\|\boldsymbol{h}_i(\boldsymbol{x}_i)\| \leqslant h_G$; ι_l and b_l are the center and bandwidth of $\boldsymbol{h}_i(\boldsymbol{x}_i)$, respectively; ε_i is the approximation error of the NN satisfying $\|\varepsilon_i\| \leqslant \varepsilon_G$; Ω is a compact set; Θ_G, h_G and ε_G are positive constants.

Let $\hat{\boldsymbol{g}}_i(\hat{\boldsymbol{x}}_i(t))$ be the estimation of $\boldsymbol{g}_i(\boldsymbol{x}_i(t))$. The expression is designed as follows:

$$\hat{\boldsymbol{g}}_i(\hat{\boldsymbol{x}}_i(t)) = \hat{\boldsymbol{\Theta}}_i^{\mathrm{T}}(t)\boldsymbol{h}_i(\hat{\boldsymbol{x}}_i) \tag{3}$$

where $\hat{\boldsymbol{\Theta}}_i(t)$ is the estimation of the weight matrix $\boldsymbol{\Theta}_i$, and the corresponding adaptive law is designed as follows:

$$\dot{\hat{\boldsymbol{\Theta}}}_i(t) = \boldsymbol{\Gamma}_{\boldsymbol{\Theta}_i}\left[\boldsymbol{h}_i(\hat{\boldsymbol{x}}_i)(\boldsymbol{y}_i - \boldsymbol{C}_i\hat{\boldsymbol{x}}_i) - \frac{\delta}{2}\hat{\boldsymbol{\Theta}}_i(t)\right] \tag{4}$$

where $\hat{\boldsymbol{x}}_i$ is the state measured using the observer designed subsequently; $\boldsymbol{\Gamma}_{\boldsymbol{\Theta}_i}$ is the positive matrix; δ is the positive constant; let $\|\boldsymbol{h}_i(\cdot)\| < h_r$ and $\|\boldsymbol{h}_{ei}(\boldsymbol{x}_i)\| = \|\boldsymbol{h}_i(\boldsymbol{x}_i) - \boldsymbol{h}_i(\hat{\boldsymbol{x}}_i)\| \leqslant \gamma < h_r$.

1.3 Memory-Based Event-Triggered Strategy

This paper adopts a memory-based event-triggered mechanism shown in Fig. 1. Compared with the event-triggered mechanisms designed in Ref. [31], not only the state information from the previous time step is considered, but also the state information from the previous m time steps is taken into account for observation.

Firstly, the notation $\boldsymbol{s}_{i,t_{k-w+1}}(t)$ is used to denote the error between the states of the previous m-time steps and that at the current time t, shown as below:

$$\boldsymbol{s}_{i,t_{k-w+1}}(t) = \boldsymbol{z}_i(t^i_{k-w+1}) - \boldsymbol{z}_i(t) \qquad w = 1,\cdots,m \tag{5}$$

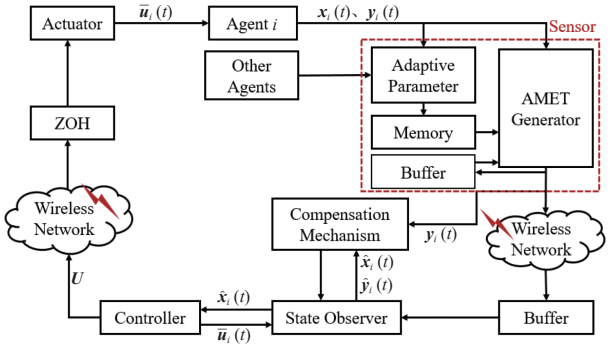


Fig. 1 System Architecture Diagram Under DoS Attack

where $\boldsymbol{z}_i(t)$ is the state error between the agent and other agents, satisfying:

$$\boldsymbol{z}_i(t)=\sum_{j=1,j\neq i}^{N}(\boldsymbol{x}_i(t)-\boldsymbol{x}_j(t)) \tag{6}$$

Then, the memory-based event-triggered condition that determines the update of the next triggered time are given as follows:

$$\begin{cases} t_{k+1}^i=\inf\{t>t_k^i \mid f_i(t)>0\} \\ f_i(t)=\left(\sum_{w=1}^{m}\varsigma_w\boldsymbol{s}_{i,t_\varrho}(t)\right)^{\mathrm{T}}\boldsymbol{\Lambda}\left(\sum_{w=1}^{m}\varsigma_w\boldsymbol{s}_{i,t_\varrho}(t)\right)- \\ \qquad \boldsymbol{\epsilon}_i(t)\boldsymbol{z}_i^{\mathrm{T}}(t)\boldsymbol{\Lambda}\boldsymbol{z}_i(t) \end{cases} \tag{7}$$

where t_k^i and t_{k+1}^i denote the kth and $(k+1)$th triggered instant of the agent i, respectively; the notation $\varrho=k-w+1$; and $\boldsymbol{\Lambda}=\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C}$; ς_w represents the weight values corresponding to the historical data at different time points used by the memory-based event-triggered mechanism and satisfies $\sum_{w=1}^{m}\varsigma_w=1$; $\boldsymbol{\epsilon}_i(t)$ is the adaptive weight of the triggered function, which satisfies the following update rate:

$$\dot{\boldsymbol{\epsilon}}_i(t)=\begin{cases}\boldsymbol{\tau}_i\boldsymbol{z}_i^{\mathrm{T}}(t)\boldsymbol{\Lambda}\boldsymbol{z}_i(t) & \boldsymbol{\epsilon}_i(t)\leqslant\bar{\epsilon}\\ 0 & \boldsymbol{\epsilon}_i(t)>\bar{\epsilon}\end{cases} \tag{8}$$

where $\boldsymbol{\tau}_i$ is the known constant; $\bar{\epsilon}$ is the upper bound of the adaptive weight matrix $\boldsymbol{\epsilon}_i(t)$ satisfying $\boldsymbol{\epsilon}_i(t)\in(0,\bar{\epsilon}]\subseteq(0,1)$.

Remark 1 In this paper, an event-triggered condition based on the cumulative error of historical data is proposed. This condition effectively controls the frequency of triggered events by accumulating the error between historical sampled outputs and the latest triggered output, thereby preventing the occurrence of triggered events infinitely within a short time interval. Consequently, the lower bound of the triggered interval $\iota=t_{k+1}-t_k>0$ is automatically guaranteed by condition Eq.(7), ensuring that the Zeno phenomenon is avoided[32].

Remark 2 The primary difference between the proposed memory-based adaptive event-triggered mechanism and the mainstream adaptive event-triggered mechanism[33-34] lies in the integration of a buffer (Fig. 1), which is used to store the most recent w triggered states. When a new triggered state $x(t_{k+1})$ is stored in the buffer, the earliest triggered state $x(t_{k+1-m})$ is removed, ensuring that only the latest w successfully triggered states are retained. These data are utilized by assigning different weights ς_w, with the most recent triggered state $x(t_{k+1})$ being assigned the biggest weight ς_1. When $m=1$, the mechanism reduces to the traditional adaptive event-triggered mechanism. By making full use of historical data, the system's dynamic changes can be monitored more accurately, leading to improved event-triggered sensitivity and reliability.

1.4 DoS Attacks

The DoS attacks, which occupy network resources by sending a large amount of useless data to the network and damaging the network, are considered (Fig. 1). The Bernoulli process is adopted to describe this situation, and the affected controller is designed as follows:

$$\bar{\boldsymbol{u}}_i(t)=r(t)\boldsymbol{u}_i(t) \tag{9}$$

where $r(t)$ is the Bernoulli distributed variable to denote the frequency of DoS attacks satisfying $\mathbb{E}\{r(t)=1\}=r_1$ with $r_1\in[0,1]$ is a known constant. Here, a switch matrix $\boldsymbol{U}$ is introduced to describe DoS attacks:

$$\boldsymbol{U}=\begin{cases}1 & \text{No attacks}\\ 0 & \text{DoS attacks}\end{cases}$$

Remark 3 If $\boldsymbol{U}$ is used directly in Eq.(9), it would result in actuator failure rather than DoS attacks, which is why the Bernoulli process is invoked. Furthermore, using the Bernoulli process provides a better way to characterize the randomness and suddenness of network resource consumption in Ref.[35].

1.5 Observer-Based Controller Design

The state information of all agents is often difficult to obtain directly in practical applications. Motivated by Ref.[27], a NN-based controller is designed in this paper to handle this problem:

$$\begin{cases}\dot{\hat{\boldsymbol{x}}}_i(t)=\boldsymbol{A}\hat{\boldsymbol{x}}_i(t)+\boldsymbol{B}\bar{\boldsymbol{u}}_i(t)+\boldsymbol{F}\hat{\boldsymbol{g}}_i(\hat{\boldsymbol{x}}_i(t))+\boldsymbol{Q}_i(t)+ \\ \qquad\qquad \boldsymbol{D}\boldsymbol{R}_i(t)+\boldsymbol{L}_\sigma\boldsymbol{C}\left[\sum_{w=1}^{m}\varsigma_w\boldsymbol{z}_i(t_\varrho^i)-\hat{\boldsymbol{z}}_i(t)\right] \\ \hat{\boldsymbol{y}}_i(t)=\boldsymbol{C}\hat{\boldsymbol{x}}_i(t)\end{cases} \tag{10}$$

where $\hat{\boldsymbol{x}}_i(t)$ is the observed state of agent i; $\boldsymbol{L}_\sigma$ is the observer gain to be designed; $\boldsymbol{Q}_i(t)$ and $\boldsymbol{R}_i(t)$ are the compensation mechanisms to achieve compensation for the uncertainties $\Delta\boldsymbol{A}$ and disturbances $d_i(t)$, as follows:

$$\boldsymbol{Q}_i(t)=$$
$$\begin{cases}\mu_1\tau^2(1+\mu_2)\dfrac{\hat{\boldsymbol{x}}_i^{\mathrm{T}}(t)\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\hat{\boldsymbol{x}}_i(t)\boldsymbol{P}_1^{-1}\boldsymbol{C}^{\mathrm{T}}\boldsymbol{s}_{y_i}(t)}{2\boldsymbol{s}_{y_i}^{\mathrm{T}}(t)\boldsymbol{s}_{y_i}(t)} & \text{if } \boldsymbol{s}_{y_i}\neq 0\\ 0 & \text{if } \boldsymbol{s}_{y_i}=0\end{cases}$$

$$R_i(t)=\begin{cases} d_r\dfrac{\boldsymbol{G}\boldsymbol{s}_{y_i}(t)}{\left\|\boldsymbol{G}\boldsymbol{s}_{y_i}(t)+\wp_1\right\|} & \text{if } \boldsymbol{s}_{y_i}\neq 0 \\ 0 & \text{if } \boldsymbol{s}_{y_i}=0 \end{cases}$$

where $\wp_1$, μ_1, τ and μ_2 are positive scalars; $\boldsymbol{s}_{y_i}$ denotes the error of the measured output $\boldsymbol{y}_i(t)$ with $\boldsymbol{s}_{y_i}=\boldsymbol{y}_i(t)-\hat{\boldsymbol{y}}_i(t)$; $\boldsymbol{P}_1$ and $\boldsymbol{G}$ are the matrices to be determined.

Considering Eqs.(1), (3) and (10), the observer-based controller will be designed as follows:

$$\begin{cases} \bar{\boldsymbol{u}}_i(t)=-\boldsymbol{U}(\boldsymbol{K}\hat{\boldsymbol{x}}_i(t)+\boldsymbol{B}^{-1}\boldsymbol{F}\hat{\boldsymbol{g}}_i(\hat{\boldsymbol{x}}_i(t))+\boldsymbol{B}^{-1}\boldsymbol{D}\boldsymbol{\Upsilon}_i(t)) \\ \boldsymbol{\Upsilon}_i(t)=\begin{cases} d_r\left(\dfrac{\boldsymbol{G}_1\boldsymbol{\zeta}_{y_i}(t)}{\|\boldsymbol{G}_1\boldsymbol{\zeta}_{y_i}(t)\|+\wp_2}\right) & \boldsymbol{\zeta}_{y_i}\neq 0 \\ 0, & \boldsymbol{\zeta}_{y_i}=0 \end{cases} \end{cases} \tag{11}$$

where $\wp_2$ is a positive scalar; $\boldsymbol{\zeta}_{y_i}(t)$ denotes the error between the output of the ith agent and the average output with $\boldsymbol{\zeta}_{y_i}(t)=\boldsymbol{y}_i(t)-\dfrac{1}{N}\sum_{i=1}^{N}\boldsymbol{y}_i(t)$; $\boldsymbol{B}^{-1}$ is the generalized inverse of $\boldsymbol{B}$, satisfying $(\boldsymbol{I}_N-\boldsymbol{B}\boldsymbol{B}^{-1})\boldsymbol{D}=0$ and $(\boldsymbol{I}_N-\boldsymbol{B}\boldsymbol{B}^{-1})\boldsymbol{F}=0$; $\boldsymbol{K}$ and $\boldsymbol{G}_1$ are the matrices to be determined.

Substituting Eq.(11) in Eq.(1), the resulted closed-loop system can be obtained as following:

$$\begin{aligned} \dot{\boldsymbol{x}}(t)=&[\boldsymbol{I}_N\otimes(\boldsymbol{A}-\boldsymbol{B}\boldsymbol{U}\boldsymbol{K}+\Delta\boldsymbol{A})]\boldsymbol{x}(t)+(\boldsymbol{I}_N\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})\boldsymbol{s}_x(t)+\\ &(\boldsymbol{I}_N\otimes\boldsymbol{F})\left[\boldsymbol{g}(\boldsymbol{x})-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\hat{\boldsymbol{g}}(\hat{\boldsymbol{x}})\right]-(\boldsymbol{I}_N\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1})\boldsymbol{\Upsilon}(t)+\\ &(\boldsymbol{I}_N\otimes\boldsymbol{D})\boldsymbol{d}(t) \end{aligned}$$

where $\boldsymbol{s}_x(t)=[s_{x_1}(t)\ \cdots\ s_{x_N}(t)]^{\mathrm{T}}$ is the estimation error of the state with $\boldsymbol{s}_{x_i}=\boldsymbol{x}_i-\hat{\boldsymbol{x}}_i$. To simplify notations, $\boldsymbol{x}(t)$, $\boldsymbol{g}(x)$, $\hat{\boldsymbol{g}}(\hat{x})$, $\boldsymbol{d}(t)$ and $\boldsymbol{\Upsilon}(t)$ are constructed in a manner similar to $\boldsymbol{s}_x(t)$. Then we have:

$$\begin{aligned} \dot{\boldsymbol{s}}_x(t)=&(\boldsymbol{I}_N\otimes\boldsymbol{A}-\boldsymbol{\mathcal{L}}\otimes\boldsymbol{L}_\sigma\boldsymbol{C})\boldsymbol{s}_x(t)+(\boldsymbol{I}_N\otimes\Delta\boldsymbol{A})\boldsymbol{x}(t)+\\ &(\boldsymbol{I}_N\otimes\boldsymbol{F})\tilde{\boldsymbol{\Theta}}(t)+(\boldsymbol{I}_N\otimes\boldsymbol{D})(\boldsymbol{d}(t)-\boldsymbol{R}(t))-\\ &(\boldsymbol{I}_N\otimes\boldsymbol{I})\boldsymbol{Q}(t)-(\boldsymbol{I}_N\otimes\boldsymbol{L}_\sigma\boldsymbol{C})\sum_{w=1}^{m}\varsigma_w\boldsymbol{s}_{t_{k-w+1}}(t) \end{aligned} \tag{12}$$

where $\tilde{\boldsymbol{\Theta}}(t)=\boldsymbol{\Theta}^{\mathrm{T}}\boldsymbol{h}(\boldsymbol{x})+\varepsilon-\hat{\boldsymbol{\Theta}}^{\mathrm{T}}(t)\boldsymbol{h}(\hat{\boldsymbol{x}})$.

Moreover, let $\zeta_{x_i}(t)$ be the error between the state of the ith agent and the average state with $\zeta_{x_i}(t)=\boldsymbol{x}_i(t)-\dfrac{1}{N}\sum_{i=1}^{N}\boldsymbol{x}_i(t)$, satisfying $\boldsymbol{\zeta}(t)=[\zeta_{x_1}(t)\cdots\zeta_{x_N}(t)]^{\mathrm{T}}=(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{I}_N)\boldsymbol{x}(t)$ with $\boldsymbol{\mathcal{M}}=\boldsymbol{I}_N-\dfrac{1}{N}\boldsymbol{1}\boldsymbol{1}^{\mathrm{T}}$ and $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{M}}=\boldsymbol{\mathcal{M}}\boldsymbol{\mathcal{L}}=\boldsymbol{\mathcal{L}}$. Then, the tracking error system is obtained as follows:

$$\begin{aligned} \dot{\boldsymbol{\zeta}}(t)=&[\boldsymbol{I}_N\otimes(\boldsymbol{A}-\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})+\Delta\boldsymbol{A}]\zeta(t)+(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})\boldsymbol{s}_x(t)+\\ &(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{D})\boldsymbol{d}(t)+(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{F})(\boldsymbol{g}(\boldsymbol{x})-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\hat{\boldsymbol{g}}(\hat{\boldsymbol{x}}))-\\ &(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{D})\boldsymbol{\Upsilon}(t) \end{aligned} \tag{13}$$

Problem Statement: Given a MIMO high-order nonlinear multi-agent system with time-varying uncertainties and extended disturbances in Eq.(1), we focus on the NN-based observer-based controller design under DoS attacks in Eq.(9), so that the closed-loop multi-agent error system in Eq.(13) is uniformly and ultimately bounded.

2 Main Results

This section presents a memory-based event-triggered method for consensus analysis and an observer-based controller design for MIMO high-order nonlinear multi-agent systems.

2.1 Consensus Analysis

Theorem 1 For given positive scalars μ_1, μ_2, μ_3, τ, ϑ_1, ϑ_2, and δ, if there exist symmetrical positive definite matrices $\boldsymbol{P}_1$, $\boldsymbol{P}_2$, matrices $\boldsymbol{L}_\sigma$, $\boldsymbol{G}$, $\boldsymbol{G}_1$, $\boldsymbol{K}$ such that:

$$\begin{bmatrix} \boldsymbol{Q}_{11} & \boldsymbol{Q}_{12} & \boldsymbol{Q}_{13} \\ * & \boldsymbol{Q}_{22} & \boldsymbol{Q}_{23} \\ * & * & \boldsymbol{Q}_{33} \end{bmatrix}<0 \tag{14}$$

$$\begin{bmatrix} -\delta\boldsymbol{I} & \boldsymbol{P}_1\boldsymbol{D}-\boldsymbol{C}^{\mathrm{T}}\boldsymbol{G} \\ * & -\delta\boldsymbol{I} \end{bmatrix}<0 \tag{15}$$

$$\begin{bmatrix} -\delta\boldsymbol{I} & \boldsymbol{P}_2\boldsymbol{D}-\boldsymbol{C}^{\mathrm{T}}\boldsymbol{G}_1 \\ * & -\delta\boldsymbol{I} \end{bmatrix}<0 \tag{16}$$

where

$$\begin{aligned} \boldsymbol{Q}_{11}=&\mathrm{He}(\boldsymbol{P}_1\boldsymbol{A})-\boldsymbol{\mathcal{L}}\otimes\mathrm{He}(\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{C})+\mu_1^{-1}\boldsymbol{P}_1^2+2\vartheta_1\boldsymbol{P}_1\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_1+\\ &\tau^2\mu_1\left(1+\frac{1}{\mu_2}\right)\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}+\mu_3^{-1}\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{L}_\sigma^{\mathrm{T}}\boldsymbol{P}_1+\pi\boldsymbol{P}_1 \end{aligned}$$

$$\begin{aligned} \boldsymbol{Q}_{22}=&\mathrm{He}(\boldsymbol{P}_2\boldsymbol{A}-\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})+\tau\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2+\boldsymbol{I})+\pi\boldsymbol{P}_2+\\ &\mu_3\bar{\epsilon}(L^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})+\lambda_{\max}(\boldsymbol{\mathcal{M}})\boldsymbol{P}_2^2+2\vartheta_2(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_2) \end{aligned}$$

$$\boldsymbol{Q}_{33}=-\delta\boldsymbol{I}+\pi\boldsymbol{P}_3,\quad \boldsymbol{Q}_{12}=-\lambda_{\max}(\boldsymbol{\mathcal{M}})\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{K}$$

$$\boldsymbol{Q}_{13}=h_r(\boldsymbol{C}-\boldsymbol{P}_1\boldsymbol{F}),\quad \boldsymbol{Q}_{23}=-\lambda_{\max}(\boldsymbol{\mathcal{M}})h_r\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}$$

Then the underlying system in Eq.(13) can achieve consensus with UUB.

Proof Considering the underlying system in Eqs. (3), (12) and (13), we select the Lyapunov function as:

$$V(t)=V_1(t)+V_2(t)+V_3(t) \tag{17}$$

where $V_1(t)=\boldsymbol{s}_x^{\mathrm{T}}(t)\boldsymbol{P}_1\boldsymbol{s}_x(t)$, $V_2(t)=\boldsymbol{\zeta}^{\mathrm{T}}(t)\boldsymbol{P}_2\boldsymbol{\zeta}(t)$, $V_3(t)=\boldsymbol{\Theta}_e^{\mathrm{T}}(t)\boldsymbol{P}_3\boldsymbol{\Theta}_e(t)$; $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$ are symmetrical positive determined matrices; $\boldsymbol{\Theta}_e(t)=\hat{\boldsymbol{\Theta}}(t)-\boldsymbol{\Theta}$ is the error between the weight matrix $\boldsymbol{\Theta}(t)$ and it's estimation; $\boldsymbol{P}_3=\boldsymbol{\Gamma}_{\boldsymbol{\Theta}}^{-1}$ and $\boldsymbol{\Gamma}_{\boldsymbol{\Theta}}$ is a known positive definite matrix. Then,

$$\begin{aligned} \dot{V}_1(t)=&2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1)\dot{\boldsymbol{s}}_x(t)\\ =&2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1)\Big\{(\boldsymbol{I}_N\otimes\boldsymbol{A}-\boldsymbol{\mathcal{L}}\otimes\boldsymbol{L}_\sigma\boldsymbol{C})\boldsymbol{s}_x(t)+\\ &(\boldsymbol{I}_N\otimes\Delta\boldsymbol{A})\boldsymbol{x}(t)+(\boldsymbol{I}_N\otimes\boldsymbol{F})\tilde{\boldsymbol{\Theta}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{D})(\boldsymbol{d}(t)-\boldsymbol{R}(t))-\\ &(\boldsymbol{I}_N\otimes\boldsymbol{I})\boldsymbol{Q}(t)-(\boldsymbol{I}_N\otimes\boldsymbol{L}_\sigma\boldsymbol{C})\sum_{w=1}^{m}\varsigma_w\boldsymbol{s}_{t_0}(t)\Big\} \end{aligned} \tag{18}$$

where $\tilde{\boldsymbol{\Theta}}(t)=\boldsymbol{\Theta}^{\mathrm{T}}\boldsymbol{h}(\boldsymbol{x})+\varepsilon-\hat{\boldsymbol{\Theta}}^{\mathrm{T}}(t)\boldsymbol{h}(\hat{\boldsymbol{x}})$. Motivated by Ref.[27], we have

$$2\boldsymbol{s}_x^{\mathrm{T}}(t)\boldsymbol{P}_1(\boldsymbol{I}_N\otimes\Delta\boldsymbol{A})\boldsymbol{x}(t)$$
$$\leqslant\frac{1}{\mu_1}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1^2)\boldsymbol{s}_x(t)+\mu_1\tau^2\left(1+\frac{1}{\mu_2}\right)\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})\boldsymbol{s}_x(t)+$$
$$\mu_1\tau^2(1+\mu_2)\hat{\boldsymbol{x}}^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})\hat{\boldsymbol{x}}(t) \tag{19}$$
$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_2\Delta\boldsymbol{A})\boldsymbol{\zeta}(t)$$
$$\leqslant\boldsymbol{\zeta}^{\mathrm{T}}(t)\left[\tau\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2+\boldsymbol{I})\right]\boldsymbol{\zeta}(t) \tag{20}$$

Based on Young's inequality and the memory-based design mechanism, we can obtain:

$$-2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{C})\sum_{w=1}^{m}\varsigma_w\boldsymbol{s}_{t_e}(t)$$
$$\leqslant\frac{1}{\mu_3}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{L}_\sigma^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+$$
$$\mu_3\epsilon(t)\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{L}}^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})\boldsymbol{\zeta}(t) \tag{21}$$

Subsequently, we consider the nonlinear part:

$$2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F})\tilde{\boldsymbol{\Theta}}(t)$$
$$\leqslant2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F})\left[\gamma\boldsymbol{\Theta}^{\mathrm{T}}(t)+\varepsilon-\boldsymbol{\Theta}_e^{\mathrm{T}}(t)\boldsymbol{h}(\hat{\boldsymbol{x}})\right]$$
$$\leqslant2\mathfrak{d}_1\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+\frac{1}{\mathfrak{d}_1}\Theta_G^2\gamma^2+\frac{1}{\mathfrak{d}_1}\varepsilon_G^2-$$
$$2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F})\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r \tag{22}$$

Considering $2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{D})(\boldsymbol{d}(t)-\boldsymbol{R}(t))\leqslant0$, we have:

$$\dot{V}_1(t)\leqslant2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{A}-\boldsymbol{\mathcal{L}}\otimes\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{C})\boldsymbol{s}_x(t)+\frac{1}{\mathfrak{d}_1}\Theta_G^2\gamma^2+$$
$$\frac{1}{\mathfrak{d}_1}\varepsilon_G^2+\frac{1}{\mu_1}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+$$
$$\mu_1\tau^2\left(1+\frac{1}{\mu_2}\right)\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})\boldsymbol{s}_x(t)+$$
$$\frac{1}{\mu_3}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{L}_\sigma^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+$$
$$\mu_3\epsilon(t)\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{L}}^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})\boldsymbol{\zeta}(t)+$$
$$2\mathfrak{d}_1\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{P}_1)\boldsymbol{s}_x(t)-$$
$$2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F})\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r \tag{23}$$

For the second part of the Lyapunov function, we have:

$$\dot{V}_2(t)=2\boldsymbol{\zeta}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_2)\dot{\boldsymbol{\zeta}}(t)$$
$$=2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_2)\left\{[\boldsymbol{I}_N\otimes(\boldsymbol{A}-\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})+\Delta\boldsymbol{A}]\boldsymbol{\zeta}(t)+\right.$$
$$(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})\boldsymbol{s}_x(t)+(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{F})[\boldsymbol{g}(x)-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\hat{\boldsymbol{g}}(\hat{x})]+$$
$$\left.(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{D})\boldsymbol{d}(t)-(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{D})\boldsymbol{\Upsilon}(t)\right\} \tag{24}$$

In the case of a secure network, *i.e.*, $\boldsymbol{U}=1$, due to Eq.(15) and Eq.(16), the following inequality can be established:

$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{D})\boldsymbol{d}(t)-2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{D})\boldsymbol{\Upsilon}(t)\leqslant0$$

But in the case considered in this paper, the system is subject to a certain probability of DoS attacks, then:

$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{D})\boldsymbol{d}(t)-2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{D})\boldsymbol{\Upsilon}(t)$$
$$<\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2^2)\boldsymbol{\zeta}(t)+(\|\boldsymbol{D}\|d_r)^2$$

To sum up,

$$\dot{V}_2(t)\leqslant2\boldsymbol{\zeta}^{\mathrm{T}}(t)\{\boldsymbol{I}_N\otimes\boldsymbol{P}_2[(\boldsymbol{A}-\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})]\}\boldsymbol{\zeta}(t)+\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2^2)\boldsymbol{\zeta}(t)+$$
$$\boldsymbol{\zeta}^{\mathrm{T}}(t)\left[\tau\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2+\boldsymbol{I})\right]\boldsymbol{\zeta}(t)+(\|\boldsymbol{D}\|d_r)^2+$$
$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})\boldsymbol{s}_x(t)+2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F})$$
$$\left[(\boldsymbol{h}(\boldsymbol{x})-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{h}(\hat{\boldsymbol{x}}))\boldsymbol{\Theta}^{\mathrm{T}}+\varepsilon-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{\Theta}_e^{\mathrm{T}}\boldsymbol{h}(\hat{\boldsymbol{x}})\right] \tag{25}$$

By Young's inequality and design for NN fitting of nonlinear functions, it can be obtained that:

$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F})\left[(\boldsymbol{h}(\boldsymbol{x})-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{h}(\hat{\boldsymbol{x}}))\boldsymbol{\Theta}^{\mathrm{T}}+\varepsilon-\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}\boldsymbol{h}(\hat{\boldsymbol{x}})\right]$$
$$\leqslant2\mathfrak{d}_2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{P}_2)\boldsymbol{\zeta}(t)+\frac{1}{\mathfrak{d}_2}\Theta_G^2h_r^2+\frac{1}{\mathfrak{d}_2}\varepsilon_G^2-$$
$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1})\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r \tag{26}$$

For the third part of the Lyapunov function,

$$\dot{V}_3(t)=2\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_3)\dot{\boldsymbol{\Theta}}_{\mathrm{e}}(t)$$
$$=2\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}(t)\boldsymbol{h}(\hat{\boldsymbol{x}})\boldsymbol{C}\boldsymbol{s}_x(t)-\delta\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}(t)\hat{\boldsymbol{\Theta}}(t)$$
$$\leqslant2\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes C)\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r+\delta\Theta_G^2-\delta\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}(t)\boldsymbol{\Theta}_{\mathrm{e}}(t) \tag{27}$$

By substituting Eqs.(23), (25)–(27), it can produce:

$$\dot{V}(t)\leqslant\boldsymbol{s}_x^{\mathrm{T}}(t)[\boldsymbol{I}_N\otimes(\boldsymbol{P}_1\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_1)-\boldsymbol{\mathcal{L}}\otimes(\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{C}+\boldsymbol{C}^{\mathrm{T}}\boldsymbol{L}_\sigma^{\mathrm{T}}\boldsymbol{P}_1)]\boldsymbol{s}_x(t)+$$
$$\frac{1}{\mu_1}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+\mu_3\epsilon(t)\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{L}}^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})\boldsymbol{\zeta}(t)+$$
$$\frac{1}{\mu_3}\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{L}_\sigma\boldsymbol{L}_\sigma^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+$$
$$2\mathfrak{d}_1\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{P}_1\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_1)\boldsymbol{s}_x(t)+d_r^2+$$
$$\mu_1\tau^2\left(1+\frac{1}{\mu_2}\right)\boldsymbol{s}_x^{\mathrm{T}}(t)(\boldsymbol{I}_N\otimes\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})\boldsymbol{s}_x(t)+$$
$$\boldsymbol{\zeta}^{\mathrm{T}}(t)[\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2-\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{K}-\boldsymbol{K}^{\mathrm{T}}\boldsymbol{U}^{\mathrm{T}}\boldsymbol{B}^{\mathrm{T}}\boldsymbol{P}_2)]\boldsymbol{\zeta}(t)+$$
$$\boldsymbol{\zeta}^{\mathrm{T}}(t)\tau\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2+\boldsymbol{I})\boldsymbol{\zeta}(t)+\delta\Theta_G^2+$$
$$2\mathfrak{d}_2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_2)\boldsymbol{\zeta}(t)+\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{P}_2)\boldsymbol{\zeta}(t)+$$
$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{B}\boldsymbol{U}\boldsymbol{K})\boldsymbol{s}_x(t)+\frac{1}{\mathfrak{d}_1}\gamma^2\Theta_G^2+\frac{1}{\mathfrak{d}_2}h_r\Theta_G^2+$$
$$\left(\frac{1}{\mathfrak{d}_1}+\frac{1}{\mathfrak{d}2}\right)\varepsilon_G^2+2\boldsymbol{s}_x^{\mathrm{T}}(t)[\boldsymbol{I}_N\otimes(\boldsymbol{C}-\boldsymbol{P}_1\boldsymbol{F})]\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r-$$
$$2\boldsymbol{\zeta}^{\mathrm{T}}(t)(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1})\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}h_r-\delta\boldsymbol{\Theta}_{\mathrm{e}}^{\mathrm{T}}(t)\boldsymbol{\Theta}_{\mathrm{e}}(t) \tag{28}$$

Let $\boldsymbol{\mho}=[\boldsymbol{s}_x(t)\ \ \boldsymbol{\zeta}(t)\ \ \boldsymbol{\Theta}_e(t)]^{\mathrm{T}}$, Eq.(14) in Theorem 1 can be obtained based on Eq.(28) using the Schur Complement lemma. Next, we can conclude that:

$$\dot{V}\leqslant-\pi V+c\boldsymbol{I} \tag{29}$$

where $c=d_r^2+\left(\frac{1}{\mathfrak{d}_1}\gamma^2+\frac{h_r}{\mathfrak{d}_2}+\delta\right)\Theta_G^2+\left(\frac{1}{\mathfrak{d}_1}+\frac{1}{\mathfrak{d}_2}\right)\varepsilon_G^2$, then $V(t)$ is bounded. By integrating both sides of Eq.(29), we obtain:

$$\int_0^t\frac{\mathrm{d}}{\mathrm{d}t}(V(\varpi)\mathrm{e}^{\pi\varpi})\,\mathrm{d}\varpi\leqslant\int_0^t c\mathrm{e}^{\pi\varpi}\mathrm{d}\varpi \tag{30}$$

$$V(t)\leqslant V(0)\mathrm{e}^{-\pi}+\frac{c}{\pi}(1-\mathrm{e}^{-\pi}) \tag{31}$$

Consequently, $V(t)<\dfrac{c}{\pi}$ as $t\to\infty$, satisfying Definition 2. This completes the proof.

2.2 Observer-Based Controller Design

In this section, Theorem 2 will be given to solve the feedback gain matrix $\boldsymbol{K}$, the compensation gain matrices $\boldsymbol{G}$ and $\boldsymbol{G}_1$, the observer gain matrix $\boldsymbol{L}$.

Theorem 2 For given positive scalars μ_1, μ_2, μ_3, $\tau>0$, ϑ_1 and ϑ_2, if there exist positive definite matrices $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$ such that the following inequality holds:

$$\begin{bmatrix}
\boldsymbol{Q}'_{11} & \boldsymbol{Q}_1 & \boldsymbol{P}_1 & \boldsymbol{P}_1\boldsymbol{F} & \boldsymbol{Q}'_{15} & 0 & 0 & 0 & \boldsymbol{Q}'_{19}\\
* & -\mu_3\boldsymbol{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
* & * & -\mu_1\boldsymbol{I} & 0 & 0 & 0 & 0 & 0 & 0\\
* & * & * & -\frac{1}{2}\vartheta_1\boldsymbol{I} & 0 & 0 & 0 & 0 & 0\\
* & * & * & * & \boldsymbol{Q}'_{55} & \boldsymbol{P}_2\boldsymbol{A} & \boldsymbol{P}_2\boldsymbol{F} & \boldsymbol{P}_2 & \boldsymbol{Q}'_{58}\\
* & * & * & * & * & -\tau\boldsymbol{I} & 0 & 0 & 0\\
* & * & * & * & * & * & -\frac{1}{2}\vartheta_2\boldsymbol{I} & 0 & 0\\
* & * & * & * & * & * & * & \boldsymbol{Q}'_{88} & 0\\
* & * & * & * & * & * & * & * & \boldsymbol{Q}'_{99}
\end{bmatrix}<0 \tag{32}$$

where

$$\begin{aligned}
\boldsymbol{Q}'_{11} &= \mathrm{He}(\boldsymbol{P}_1\boldsymbol{A})+\mu_1\tau^2\left(1+\frac{1}{\mu_2}\right)\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}+\pi\boldsymbol{P}_1-\\
&\quad \lambda_2\otimes\mathrm{He}(\boldsymbol{\Psi}\boldsymbol{C}),\\
\boldsymbol{Q}'_{55} &= \mathrm{He}(\boldsymbol{P}_2\boldsymbol{A}-\boldsymbol{B}\check{\boldsymbol{K}}\boldsymbol{U})+\tau\boldsymbol{I}+\mu_3\bar{\epsilon}(\lambda_N^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})+\pi\boldsymbol{P}_2,\\
\boldsymbol{Q}'_{15} &= \lambda_{\max}(\boldsymbol{\mathcal{M}})\check{\boldsymbol{K}}^{\mathrm{T}}\boldsymbol{U}\boldsymbol{B}^{\mathrm{T}},\\
\boldsymbol{Q}'_{19} &= h_r(\boldsymbol{C}-\boldsymbol{P}_1\boldsymbol{F}),\\
\boldsymbol{Q}'_{58} &= -\lambda_{\max}(\boldsymbol{\mathcal{M}})\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}h_r,\\
\boldsymbol{Q}'_{88} &= -\mathrm{inv}\,(\lambda_{\max}(\boldsymbol{\mathcal{M}})),\\
\boldsymbol{Q}'_{99} &= -\delta\boldsymbol{I}+\pi\boldsymbol{P}_3.
\end{aligned}$$

then the underlying system in Eq.(13) is said to be UUB, and matrices $\boldsymbol{L}$, $\boldsymbol{G}$, $\boldsymbol{G}_1$, $\boldsymbol{K}$ can be solved.

Proof Assume $\boldsymbol{\Psi}=\boldsymbol{P}_1\boldsymbol{L}_\sigma$, $\boldsymbol{P}_2\boldsymbol{B}=\boldsymbol{B}\boldsymbol{S}$, $\boldsymbol{S}\boldsymbol{K}=\check{\boldsymbol{K}}$, Eq.(14) can be rewritten as:

$$\begin{bmatrix}
\check{\boldsymbol{Q}}_{11} & \check{\boldsymbol{Q}}_{12} & \check{\boldsymbol{Q}}_{13}\\
* & \check{\boldsymbol{Q}}_{22} & \check{\boldsymbol{Q}}_{23}\\
* & * & \check{\boldsymbol{Q}}_{33}
\end{bmatrix}<0 \tag{33}$$

where

$$\begin{cases}
\check{\boldsymbol{Q}}_{11} = \mathrm{He}(\boldsymbol{P}_1\boldsymbol{A})-\boldsymbol{\mathcal{L}}\otimes\mathrm{He}(\boldsymbol{\Psi}\boldsymbol{C})+\dfrac{1}{\mu_1}\boldsymbol{P}_1^{\mathrm{T}}\boldsymbol{P}_1+\dfrac{1}{\mu_3}\boldsymbol{\Psi}^{\mathrm{T}}\boldsymbol{\Psi}+\\
\qquad \tau^2\mu_1\left(1+\dfrac{1}{\mu_2}\right)\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}+2\varepsilon_1\boldsymbol{P}_1\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_1+\pi\boldsymbol{P}_1\\
\check{\boldsymbol{Q}}_{22} = \mathrm{He}(\boldsymbol{P}_2\boldsymbol{A}-\boldsymbol{B}\check{\boldsymbol{K}}\boldsymbol{U})+\tau\boldsymbol{I}_N\otimes(\boldsymbol{P}_2\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_2+\boldsymbol{I})+\\
\qquad (\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2^{\mathrm{T}}\boldsymbol{P}_2)+\mu_3\bar{\epsilon}(\boldsymbol{L}^2\otimes\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C})+\\
\qquad 2\varepsilon_2(\boldsymbol{\mathcal{M}}\otimes\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{P}_2)+\pi\boldsymbol{P}_2\\
\check{\boldsymbol{Q}}_{33} = -\delta\boldsymbol{I}+\pi\boldsymbol{P}_3,\quad \check{\boldsymbol{Q}}_{12} = -\lambda_{\max}(\boldsymbol{\mathcal{M}})\boldsymbol{B}\check{\boldsymbol{K}}\boldsymbol{U}\\
\check{\boldsymbol{Q}}_{13} = h_r(\boldsymbol{C}-\boldsymbol{P}_1\boldsymbol{F}),\quad \check{\boldsymbol{Q}}_{23} = -\lambda_{\max}(\boldsymbol{\mathcal{M}})\boldsymbol{P}_2\boldsymbol{F}\boldsymbol{B}\boldsymbol{U}\boldsymbol{B}^{-1}h_r
\end{cases}$$

According to Eq.(14) in Theorem 1, Eq.(32) can be obtained using the Schur Complement lemma. This completes the proof.

3 Simulation Example

Motivated by Ref.[36], the effectiveness of the proposed control method will be verified on a fleet system including five unmanned vehicles and a multi-agent communication topology (Fig.2).

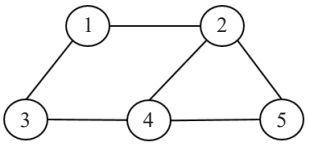


Fig. 2 The Topology Diagram of the Multi-agent Network

The vehicle model of the vehicle coordinate system is shown in Fig.3. What should be mentioned is that the longitudinal vehicle speed v_x remains constant, and only the motion of the lateral and yaw degrees of freedom will be changed by controlling the steering angles δ of the front and rear wheels. That is, the considered vehicle model can be regarded as a two-degree-of-freedom model, and the corresponding dynamic differential equations including the lateral and yaw components are formulated as follows :

$$\begin{cases}
m(\dot{y}+v_x\dot{\psi}) = F_{sf}\cos\delta_f+F_{sr}\cos\delta_r\\
I_z\ddot{\psi} = l_fF_{sf}\cos\delta_f-l_rF_{sr}\cos\delta_r
\end{cases} \tag{34}$$

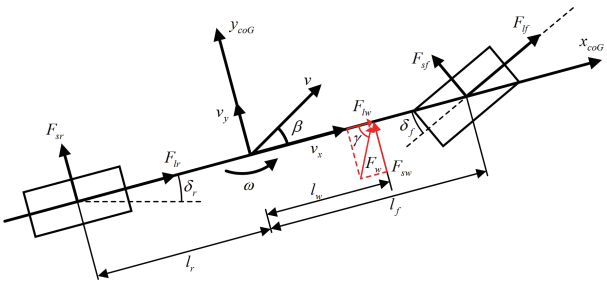


Fig. 3 The Vehicle Model of the Vehicle Coordinate System

Let us define $\boldsymbol{x}=[y\ \psi]^{\mathrm{T}}$ and $\boldsymbol{u}=[\delta_f\ \delta_r]^{\mathrm{T}}$, we can obtain the following dynamic model:

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}+\boldsymbol{B}\boldsymbol{u}+\boldsymbol{D}d_w+\boldsymbol{F}\boldsymbol{g}(x_i,t)$$

where

$$\boldsymbol{A}=\begin{bmatrix}
-\dfrac{\boldsymbol{C}_{\alpha f}+\boldsymbol{C}_{\alpha r}}{mv_x} & -1-\dfrac{\boldsymbol{C}_{\alpha f}l_f-\boldsymbol{C}_{\alpha r}l_r}{mv_x^2}\\
\dfrac{\boldsymbol{C}_{\alpha r}l_r-\boldsymbol{C}_{\alpha f}l_f}{I_zv_x} & -\dfrac{\boldsymbol{C}_{\alpha f}l_f^2+\boldsymbol{C}_{\alpha r}l_r^2}{I_zv_x}
\end{bmatrix},\boldsymbol{F}=0.01I_4,$$

$$\boldsymbol{B}=\begin{bmatrix}
\dfrac{\boldsymbol{C}_{\alpha f}}{mv_x} & \dfrac{\boldsymbol{C}_{\alpha r}}{mv_x}\\
l_f\boldsymbol{C}_{\alpha f} & -\dfrac{l_r\boldsymbol{C}_{\alpha r}}{I_z}
\end{bmatrix},D=\begin{bmatrix}
\dfrac{\sin\gamma}{m}\\
\dfrac{\sin\gamma l_w}{I_z}
\end{bmatrix},d_w=1500\sin(t)$$

and $\boldsymbol{g}(x_i,t)=[\sin(x_1)\ 0.5\sqrt{|x_1x_2|}]^{\mathrm{T}}$ is the nonlinear terms inevitably exist in the actual systems. Additionally, the remaining parameters are given in Table 1.

Table 1 The Vehicle Model Parameters

Parameter Name	Parameter Value	Unit
m	1 500	kg
I_z	2 500	kg·m^2
l_f	1.3	m
l_r	1.4	m
C_f, C_r	45 000	N/rad

The main object of this paper is to design a controller of the form of Eq.(11) under DoS attacks, such that the resulted system in Eq.(13) is UUB, using the presented result in Theorem 2. Specifically, we consider, $\tau=0.040\,4$, $\delta=0.8$, $\mu_1=4.5$, $\mu_2=0.01$, $\mu_3=0.01$, $\vartheta_1=0.3$, $\vartheta_2=0.6$, $h_r=1.5$ and $\pi=0.25$. The maximum and minimum eigenvalues of the Lyapunov matrix are $\lambda_2=1.382$ and $\lambda_N=4.618$. Then the relevant parameter matrices are solved as:

$$\boldsymbol{L}_\sigma=\begin{bmatrix}-0.001\,1 & 0.000\,2\\ 0.000\,1 & 0.001\,4\end{bmatrix},\boldsymbol{K}=\begin{bmatrix}0.013\,7 & 0.000\,8\\ 0.012\,7 & -0.000\,3\end{bmatrix},$$

$$\boldsymbol{P}_1=\begin{bmatrix}11.909\,4 & -1.508\,2\\ -1.508\,2 & 9.514\,9\end{bmatrix},\boldsymbol{P}_2=\begin{bmatrix}1.910\,2 & 0\\ 0 & 1.910\,6\end{bmatrix},$$

$$\boldsymbol{P}_3=\begin{bmatrix}0.2 & 0\\ 0 & 0.2\end{bmatrix},\boldsymbol{G}=\begin{bmatrix}11.593\,9\\ 0.202\,6\end{bmatrix},\boldsymbol{G}_1=\begin{bmatrix}1.902\,9\\ 0.342\,4\end{bmatrix}$$

Set the initial conditions as $\boldsymbol{x}_1=[0.9\ 0.5]^{\mathrm{T}}$, $\hat{\boldsymbol{x}}_1=[0.75\ 0.41]^{\mathrm{T}}$, $\boldsymbol{x}_2=[0.8\ 0.7]^{\mathrm{T}}$, $\hat{\boldsymbol{x}}_2=[0.7\ 0.66]^{\mathrm{T}}$, $\boldsymbol{x}_3=[-0.5\ -0.4]^{\mathrm{T}}$, $\hat{\boldsymbol{x}}_3=[-0.45\ -0.5]^{\mathrm{T}}$, $\boldsymbol{x}_4=[1.25\ 0.5]^{\mathrm{T}}$, $\hat{\boldsymbol{x}}_4=[1.11\ 0.58]^{\mathrm{T}}$, $\boldsymbol{x}_5=[-0.6\ -0.2]^{\mathrm{T}}$, $\hat{\boldsymbol{x}}_5=[-0.5\ -0.12]^{\mathrm{T}}$ and $v_x=30$ km/h. Subsequently, the simulation in MATLAB shows that under different lateral speeds and vehicle yaw rate derivatives, and with a $\gamma=95°$ crosswind disturbance acting on the vehicles during operation, the method designed in this paper enables the states of these 5 agents to converge (Fig. 4). Fig. 5 shows the variation curve of the adaptive rate of the NN.

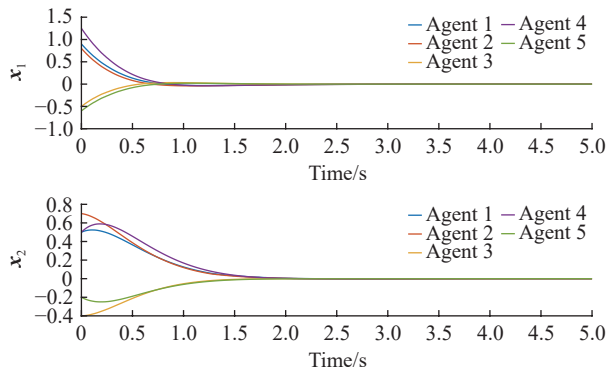


Fig. 4 System State Trajectory

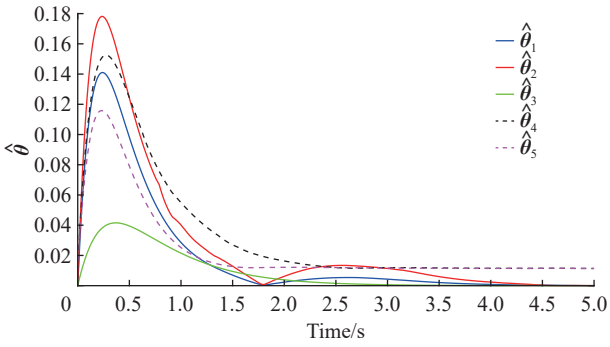


Fig. 5 The Trajectory of the Adaptive Law $\hat{\boldsymbol{\theta}}_i(t)$

Next, we compare the NN-based approach for approximating nonlinearity presented in this paper with the method in Ref.[27] that uses the Lipschitz condition to handle nonlinearity. From the curves in Fig. 6a, Fig. 6b, Fig. 7a and Fig. 7b, it can be seen that the observer-based errors in Fig. 6a and Fig. 7a are smaller than that in Fig. 6b and Fig. 7b. This indicates that the method proposed in this paper is more effective in estimating the nonlinear state compared to the method in Ref.[27], ensuring better system stability and accuracy. By setting $\bar{\epsilon}=0.3$, $\boldsymbol{\epsilon}=[0.000\,2\ \ 0.000\,15\ \ 0.000\,14\ \ 0.000\,1\ \ 0.000\,12]^{\mathrm{T}}$, $\tilde{\tau}=0.000\,1$, $m=5$, $\boldsymbol{\varsigma}_w=[0.75\ 0.12\ 0.08\ 0.04\ 0.01]^{\mathrm{T}}$. Through the event-triggered mechanism in Eq.(7), a portion of the system's resources can be saved, helping to reduce the throughput consumption of the vehicle's wireless communication system (Fig. 10 and Table 2), where $tt=\dfrac{500-\text{event triggers}}{500}$. Meanwhile, as further shown in Table 2, the method designed in this paper is more effective than the one in Ref.[27] in reducing redundant calculations and unnecessary data transmission in System (1) while

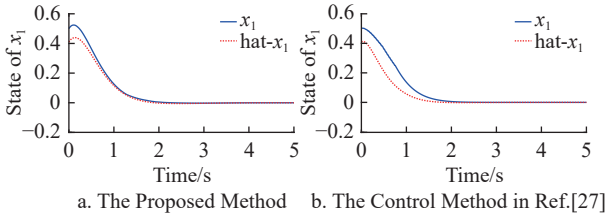


Fig. 6 The Trajectory of Agent 1's v_y and $\hat{v}_y$

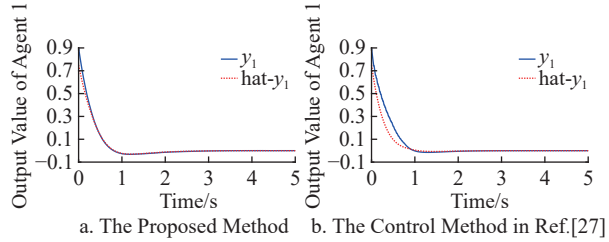


Fig. 7 The Trajectory of Agent 1's y and $\hat{y}$

Table 2 Number of Event Triggers

Agent	Date	Savings Rate *tt*/%
Agent 1	356	28.8
Agent 2	352	29.6
Agent 3	436	12.8
Agent 4	384	23.2
Agent 5	429	14.2

ensuring system performance, thereby further improving resource utilization efficiency.

Finally, a DoS attack following the Bernoulli distribution with a parameter value of $r(t) = 0.9$ will launch attacks on the system at arbitrary times. Fig. 8 and Fig. 9 show the variation in the controller input and the error between the observer-measured state and the true system state when a DoS attack occurs. Although the error $\zeta_{x_i}(t)$'s variation is affected, the system can still recover stability using the method proposed in this paper.

4 Conclusion

In this paper, for the complex disturbances, uncertainties, and network security issues faced by multi-agent systems in complex environments, a state feedback control based on observers and a compensation mechanism is proposed. Unlike most of the literature, this paper uses an adaptive NN-based method to approximate nonlinear functions, enabling its integration with LMIs. Additionally, a memory-based event-triggered mechanism is employed, which not only helps to save system resources but also, when the system is subjected to cyber attacks, combines with historical data to enhance the system's stability. Finally, the effectiveness of the proposed method is validated using a two-degree-of-freedom vehicle model. In the future, we will build upon the research in this paper to further explore the integration of the memory-based event-triggered mechanism and NNs to

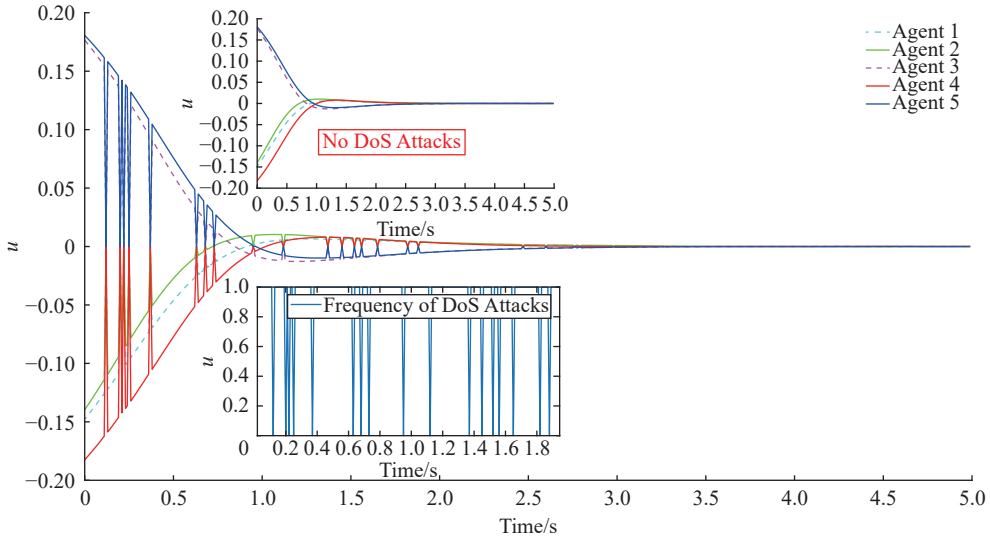


Fig. 8 Input Trajectory Under DoS Attacks

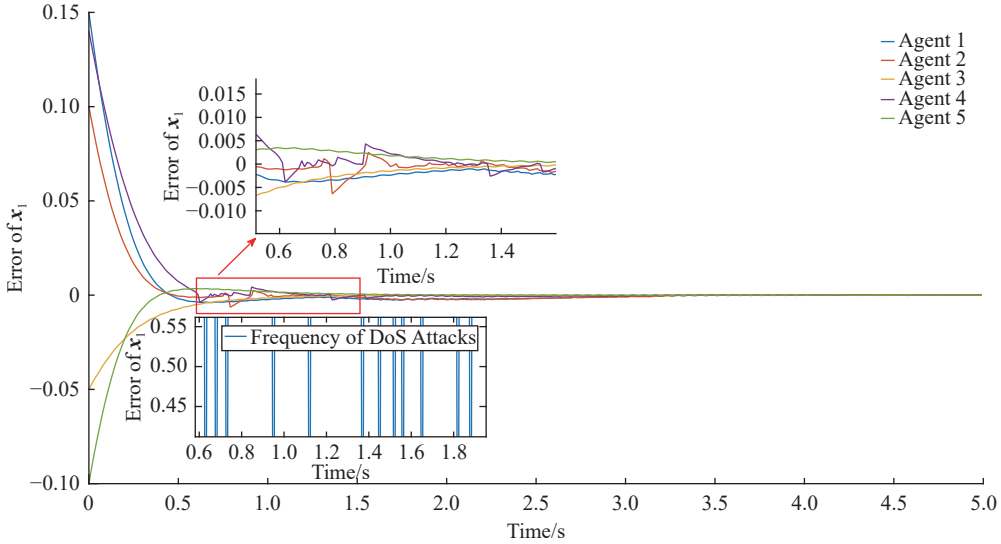


Fig. 9 Error Trajectory of x_1 Under DoS Attacks

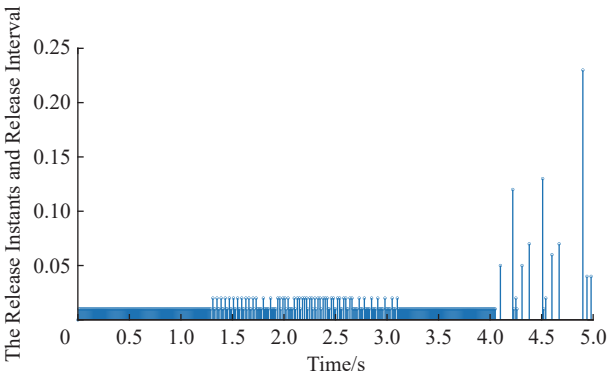


Fig. 10 Number of Event Triggers of Agent 1

address the challenges of network attacks on multi-agent systems in complex environments.

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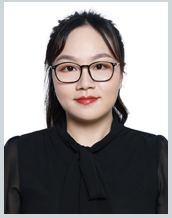
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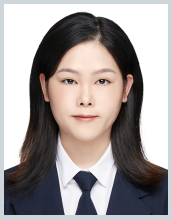
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